

20. Construct a C program to simulate Reader-Writer problem using semaphores

AIM :

To construct a C program to simulate Reader-Writer problem using semaphores

PROGRAM:

```
#include <stdio.h>
#include <pthread.h>
#include <semaphore.h>

sem_t mutex, writeBlock;

int data = 0;
int readersCount = 0;

void *reader(void *arg)
{
    int i;
    for (i = 0; i < 10; i++)
    {
        sem_wait(&mutex);
        readersCount++;
        if (readersCount == 1)
            sem_wait(&writeBlock);
        sem_post(&mutex);
        // Reading operation
        printf("Reader reads data: %d\n", data);
        sem_wait(&mutex);
        readersCount--;
        if (readersCount == 0)
            sem_post(&writeBlock);
        sem_post(&mutex);
    }
}
```

```

    return NULL;
}

void *writer(void *arg)
{
    int i;
    for (i = 0; i < 10; i++)
    {
        sem_wait(&writeBlock);

        // Writing operation

        data++;

        printf("Writer writes data: %d\n", data);

        sem_post(&writeBlock);
    }
    return NULL;
}

int main()
{
    pthread_t readers, writers;
    sem_init(&mutex, 0, 1);
    sem_init(&writeBlock, 0, 1);
    pthread_create(&readers, NULL, reader, NULL);
    pthread_create(&writers, NULL, writer, NULL);
    pthread_join(readers, NULL);
    pthread_join(writers, NULL);
    sem_destroy(&mutex);
    sem_destroy(&writeBlock);
    return 0;
}

```

OUTPUT:

Output

Status : Successfully executed

Time:

0.0000 secs

Memory:

1.732 Mb

Your Output

Writer writes data: 1

Writer writes data: 2

Writer writes data: 3

Writer writes data: 4

Writer writes data: 5

Writer writes data: 6

Writer writes data: 7

Writer writes data: 8

Writer writes data: 9

Writer writes data: 10

[illegible]